

Basic Probability and Statistics

Experiment:

An experiment is the process whose outcome is known with certainty.

Sample Space:

The set of all possible outcomes of an experiment is called sample space, and denoted by S .

Sample points:

The outcomes themselves are called sample points in the sample space.

A Random Variable:

A random variable is a function or (rule) that assigns a real number (any number greater than $-\alpha$ and less than α) to each point in the sample space S .

Example: Arnik

Denoted by X , value of random variable x .

distribution function: / cumulative distribution function:

Probability distribution : / Distribution : Giving set of

Possible values of random variable X and its probability value

$X=x$	0	1	2
$f(x)=p(x)$	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{1}{4}$

Tossing two coins
no of tails

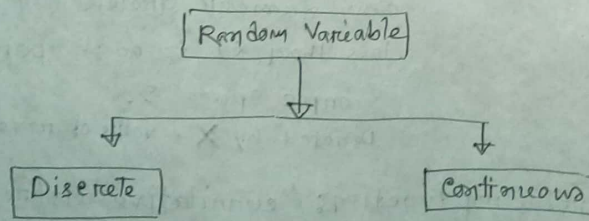
Anik

Example: Tossing two coins

$$S = \{HH, HT, TH, TT\}$$

Consider Random variable X -which gives no of heads

$$X = \{0, 1, 2\}$$



Discrete Random variable:

A Random variable x which contains a finite number of discrete values in an interval is called discrete random variable.

Example: Toss three coins

Random variable x = Number of Heads

$X = x$	0	1	2	3	→ discrete
$f(x) = P(X=x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$	

Discrete Random variable probability function/ Probability Mass Function.

$f(x)$ is probability mass function (PMF) if it satisfy two condition.

1. $f(x) \geq 0$ for each real number x
2. $\sum_{\text{all } x} f(x) = 1$.

Cumulative Distribution Function:

$F(x)$ of discrete random variable X

$$F(x) = P(X \leq x) = \sum_{x_i \leq x} f(x_i)$$

for $-\infty < x < \infty$

A distribution function $F(x)$ has following properties

1. $0 \leq F(x) \leq 1$ for all x

2. $F(x)$ is nondecreasing [if $x_1 < x_2$ then $F(x_1) \leq F(x_2)$]

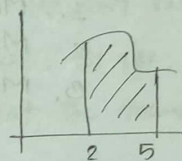
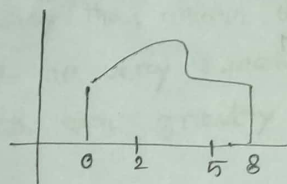
3. $\lim_{x \rightarrow \infty} F(x) = 1$ and $\lim_{x \rightarrow -\infty} F(x) = 0$

Continuous Random Variable :

A continuous random variable is a type of random variable that can take on any value within a certain interval or range.

Probability Density Function :

Take $y = f(x)$



$$\int_2^5 f(x) dx$$

1. Non negative $f(x)$.

2. $\int_0^5 f(x) dx = 1$

only the functions which satisfy the condition can be used in continuous probability distribution/ probability density function

Measure of Central Tendency: The central tendency is a measure that represents the center point of the distribution.

Distribution: A distribution is simply a collection of data and arranged from smallest to largest.

Central Tendency.

1. Mean
2. Median
3. Mode

Mean:

The mean is one of the measures of central tendency in the sense that it is the center of gravity.

Mean μ_i / Expected value $E(x_i)$

$$\mu_i = \begin{cases} \sum_{j=1}^n x_j P_{x_j}(x_j) & x \text{ is discrete} \\ \int_{-a}^a x f_{x_i}(x) dx & x \text{ continuous} \end{cases}$$

Median:

Median is the alternative measure of central tendency. It is the middle value when the data set is arranged in ascending or descending order.

The Median Maybe better Measurement:

The median Maybe better measurement of central tendency than mean when X_i can take on very large or very small values, since extreme values can greatly affect the mean.

☐ Consider a discrete random variable X and the values are 1, 2, 3, 4, 5 with probability 0.2. For this case mean and median is 3.

☐ Consider random variable Y and values are 1, 2, 3, 4 and 100. with the probability 0.2. For this case mean and median is 22 and 3.

Mode:

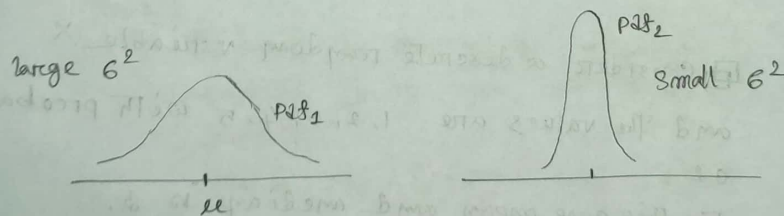
Is defined to be that value of x that maximizes

Variance:

The variance is the measure of the dispersion of a random variable about its mean. (σ^2)

$$\sigma^2 = \sum_{\text{all } x} (x - \mu)^2$$

Density function for continuous random variables with large and small variances.



The variance has following property.

1. $\text{Var}(X) \geq 0$
2. $\text{Var}(cX) = c^2 \text{Var}(X)$
3. $\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i)$

The standard deviation:

$$\sigma_i = \sqrt{\sigma_i^2}$$

Suppose mean = μ and standard deviation σ_i in normal distribution.

The probability that X_i between $\mu_i - 1.96 \sigma_i$ and $\mu_i + 1.96 \sigma_i$

$$P(\mu - 1.96\sigma < X < \mu + 1.96\sigma) = 0.95$$

$$P\left(\frac{\mu - 1.96\sigma - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{\mu + 1.96\sigma - \mu}{\sigma}\right) = 0.95$$

Stochastic process:

A stochastic process is a collection of "similar" random variables ordered over time, which are all defined on a common sample space.

1. Discrete-time stochastic process
2. Continuous-time stochastic process

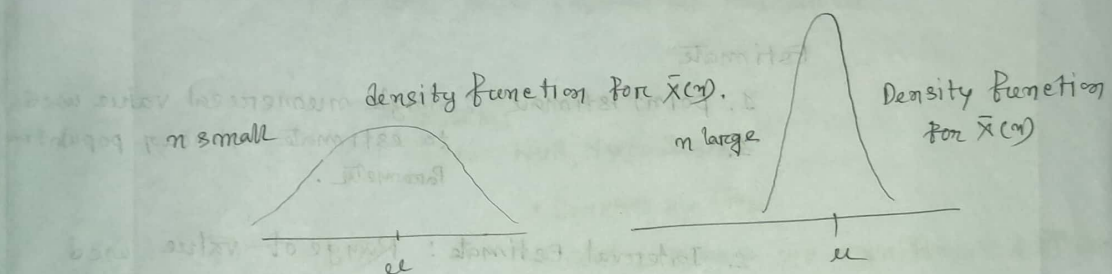
Population Mean:

$$\bar{X}(n) = \frac{\sum_{i=1}^n X_i}{n}$$

Sample variance

$$s^2(n) = \frac{\sum_{i=1}^n [X_i - \bar{X}(n)]^2}{n-1}$$

Distribution of $\bar{X}(n)$ for small and large n .



$$\left(\frac{\sigma}{\sqrt{n}}\right)$$

Estimate:

Is specific observed numerical value used to estimate an unknown population parameter.

Estimate

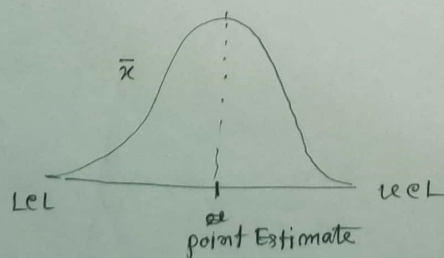
1. point estimate : single numerical value used to estimate unknown population parameter.

2. Interval Estimate : Range of value used and it is also called confidential interval

Confidential Interval.

Made of two elements

$$\begin{array}{ccc} \bar{x} + z * \left(\frac{s}{\sqrt{n}} \right) \\ \uparrow \qquad \qquad \downarrow \\ \text{Point estimate} \qquad \text{Margin Error.} \end{array}$$



95% confidence interval means the range of value that can be 95% certain or confident will contain the true mean of population.

Hypothesis Testing:

Is an educated guess to detect significant differences by comparing sample statistics with population parameter.

Hypothesis

1. Null Hypothesis:

Ex

50/50

Head / Tail

$H_0: P = 0.5$
fair coin

- Denoted by (H_0)
- States there is no significant difference
- Refer to status Quo
- Always contains the '=' sign

$$H_0: \mu_1 = \mu_2$$

2. Alternate Hypothesis:

$H_1: P \neq 0.5$

- Denoted by (H_1) and (H_a)
- H_1 always contradicts the H_0
- Challenges the status Quo
- Never contains '=' sign

$$H_1: \mu_1 \neq \mu_2$$

Error:

1. Type 1 Error:

If one rejects H_0 when in fact it is true, this is called a type 1 error, α error.

2. Type II Error:

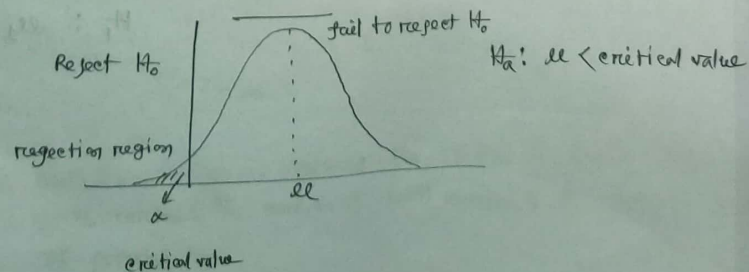
If one fails to reject H_0 when it is false, this is called a type 2 error, β error.

Types of Hypothesis test:

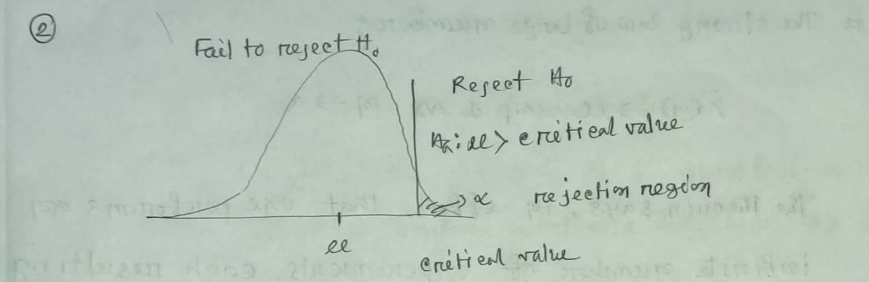
1. Left Tailed
2. Right Tailed
3. Two tailed test.

①

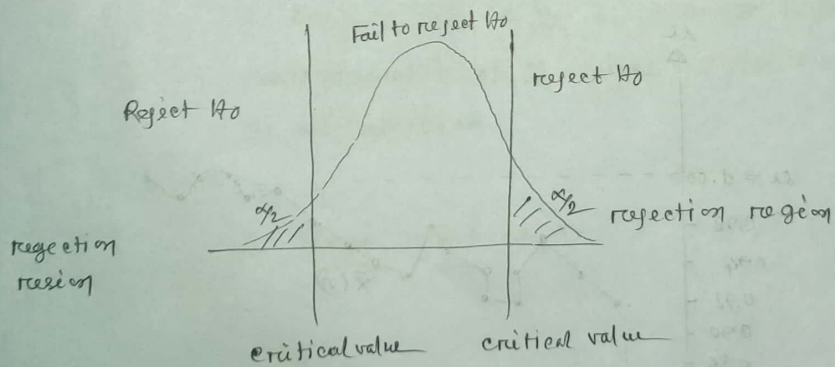
✓ Tailed: End of the distribution curve



②



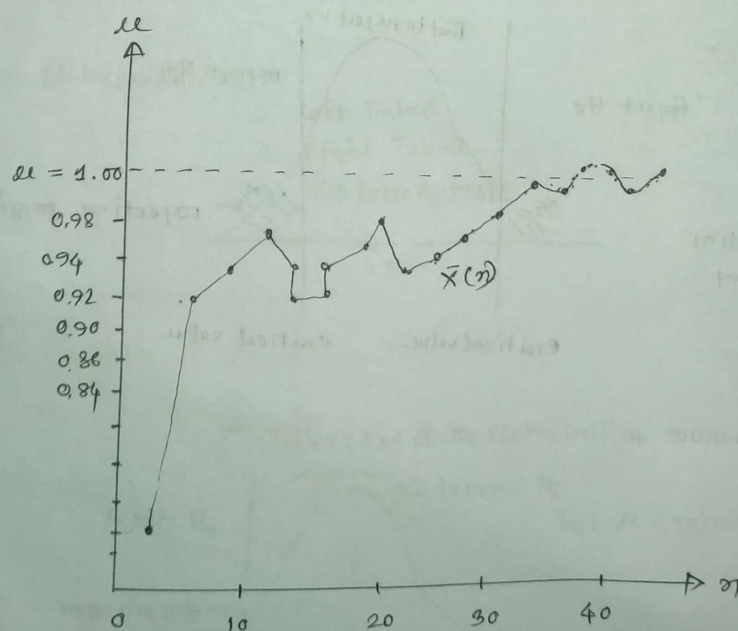
③



The strong law of large numbers:

$$\bar{X}(n) \rightarrow \mu \text{ w.p. 1 as } n \rightarrow \infty$$

The theorem says, in effect, that one performs an infinite number of experiments, each resulting in an $\bar{X}(n)$, and n is sufficiently large, then $\bar{X}(n)$ will be arbitrarily close to μ for almost all the experiments.



Parameterization of continuous distribution:

Three basic type of parameter.

1. location: defined as γ specifies an abscissa (or axis) locating point of a distribution's range of value.

usually γ is the mid point. (The mean, for example, on normal distribution.)

2. Scale: define β determine scale of measurement of the values in the range of the distribution.

3. shape: define α

Continuous Distributions:

Uniform $u(a, b)$

$$\text{Density } f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

$$\text{Distribution } F(x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & \text{if } a \leq x \leq b \\ 1 & \text{if } b < x \end{cases}$$

parameter a and b real numbers with $a < b$;

a is location parameter.

$b-a$ is scale parameter.

Range $[a, b]$

mean $\frac{a+b}{2}$

Variance $\frac{(b-a)^2}{12}$

Exponential $\text{expo}(\beta)$.

Density $f(x) = \begin{cases} \frac{1}{\beta} e^{-x/\beta} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$

Distribution $F(x) = \begin{cases} 1 - e^{-x/\beta} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$

Parameter scale parameter $\beta > 0$

Range $[0, \infty)$

Mean β

Variance β^2

Mode 0

Normal $N(\mu, \sigma^2)$.

$$\text{Density } f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}$$

Distribution No closed form

Parameters location parameter $\mu \in (-\infty, \infty)$
scale parameter. $\sigma^2 > 0$

Range $(-\infty, \infty)$

Mean μ

Variance σ^2

Mode μ

Linear Congruential Generator (LCG)

$$z_i = (a z_{i-1} + e) \bmod m$$

where

m (the modulus),

a (the multiplier),

e (the increment) and

z_0 (the seed) are non negative integers

$$m=16, a=5, e=3 \text{ and } z_0=7$$

i	$z_i = (a z_{i-1} + e) \bmod m$	$u_i = z_i/m$
0	7	—
1	6	0.375
2	1	0.0625
3	8	0.5
4	11	0.6875

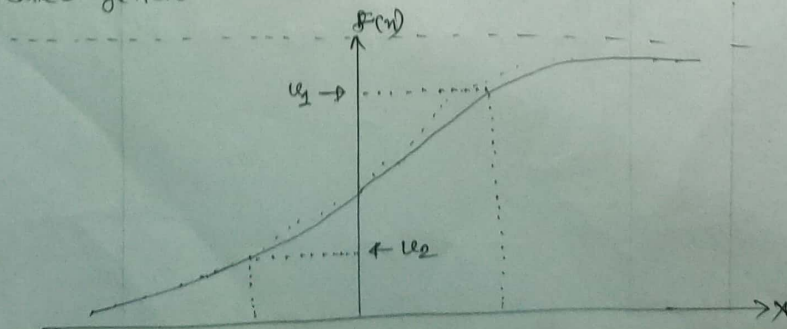
Inverse Transform:

1. Generate $u \sim U(0,1)$
2. return $X = F^{-1}(u)$

Note that $F^{-1}(u)$ will always be defined since $0 \leq u \leq 1$ and the range of F is $[0,1]$. Figure illustrates the algorithm graphically, where random variable corresponding to this distribution function can take on either positive or negative value.

The random ^{number} variable u_1 results in the positive random variate x_1 , and random number u_2 leads the negative variate x_2 .

To show that value X returned by the above algo. called general inverse transform method



Normal:

First note that given $X \sim N(0,1)$, we can obtain

$X' \sim N(\mu, \sigma^2)$ by setting $X' = \mu + \sigma \cdot X$

The polar Method:

generate $N(0,1)$ random variates in pairs is as follow

1. Generate u_1 and u_2 as IID let

$$v_i = 2u_i - 1 \quad \text{for } i = 1, 2;$$

$$w = v_1^2 + v_2^2$$

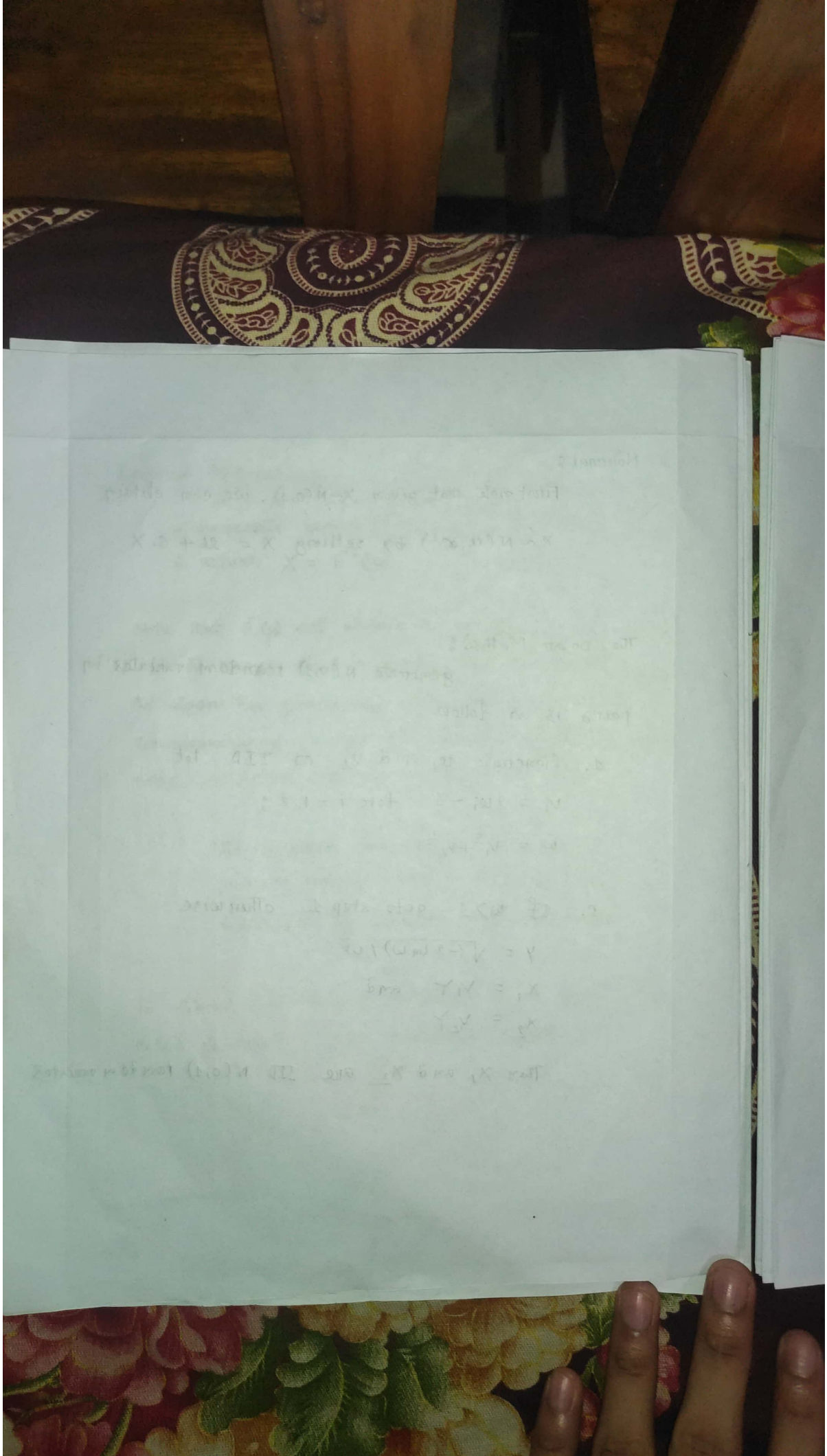
2. If $w > 1$ goto step 1 otherwise

$$y = \sqrt{(-2 \ln w) / w}$$

$$x_1 = v_1 y \quad \text{and}$$

$$x_2 = v_2 y$$

Then x_1 and x_2 are IID $N(0,1)$ random variates



Analytic Solution:

If the relationship that compose that model are simple enough, it may be possible to use mathematical method to obtain exact information on question of interest. This called an analytic solution.

Application Areas for simulation:

1. Designing and Analyzing manufacturing system
2. Evaluating military weapons system.
3. Determining Hardware requirements for communication network.
4. Determining Hardware and software requirements for a computing system.
5. Designing and operating transportation system.
6. Reengineering for business process
7. Analyzing supply chain
8. Analyzing mining operation.

Several Impediments, problems, to event wider acceptance and usefulness of simulation.

First: Models use to study large-scale system tend to be complex, and writing computer program to execute them can be difficult.

Solⁿ: This task has been made much easier in recent years by the development of excellent software products that automatically provide many of the features needed to "program" a simulation model.

Second: simulation of ^{complex} computer system is that a large amount of computer time is sometime required.

Solⁿ: This difficulty has become much less severe as computers become faster and cheaper.

Finally: There appears to be an unfortunate impression that simulation is just an exercise in computer programming, albeit a complicated one.

Soln: Consequently many simulation model "studies" have been composed of hasty model building, programming to obtain the answer. Need design and analysis of simulation models, simulation methodology carefully.

Systems:

A system is defined to be a collection of entities.

— by Schmidt and Taylor

Category of system: 1 Discrete 2 Continuous

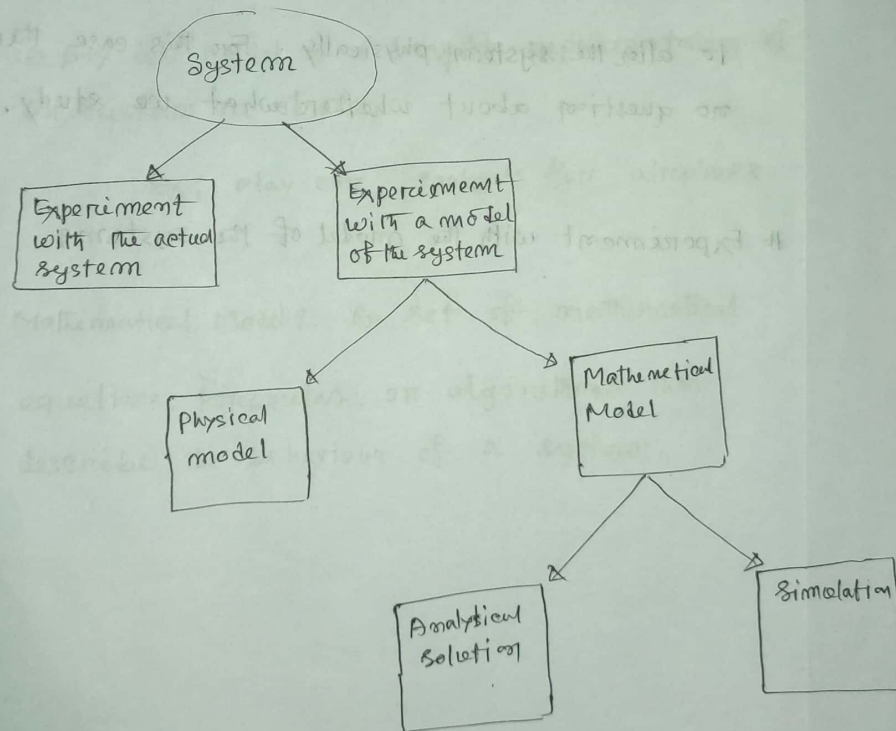
Discrete: A discrete system is one for which the state variable change instantaneously at separated points in time.

Ex: A bank

Continuous: A continuous system is one for which the state variable change continuously/over time /with respect to time.

Ex: An airplane moving.

way to study a system?



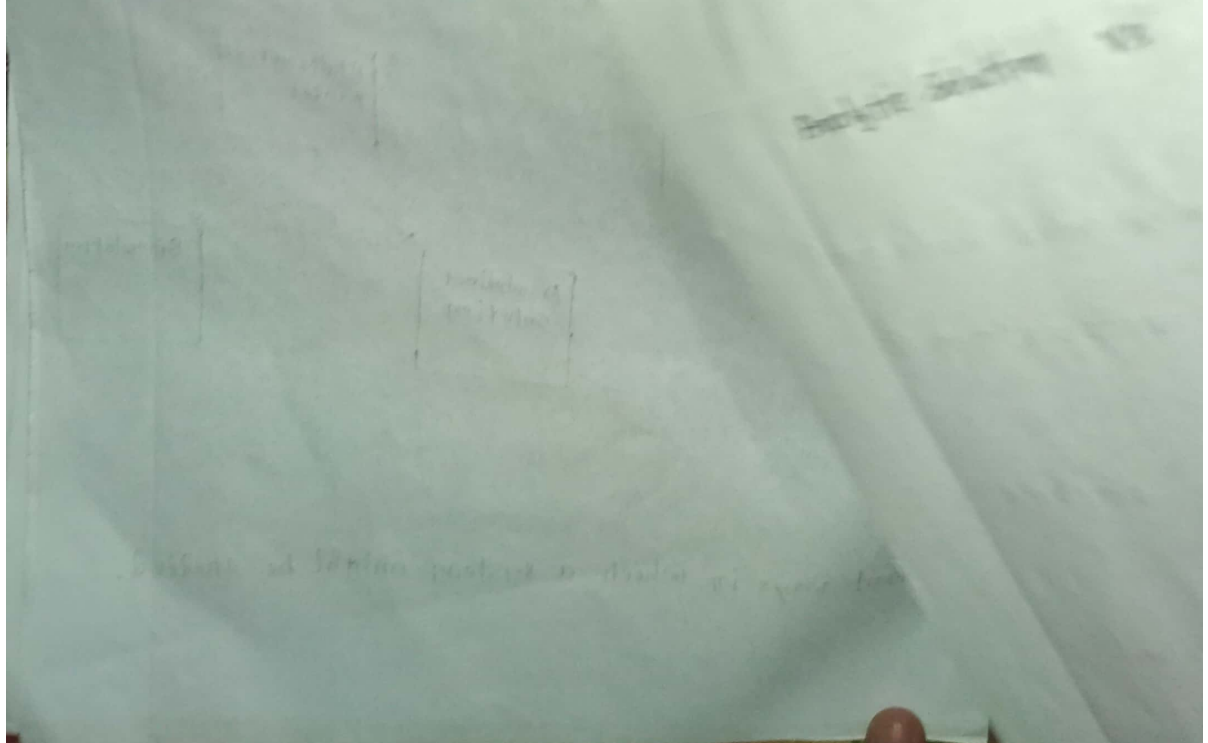
Different ways in which a system might be studied.

Experiment with the Actual System Vs De with the model of the

Experiment with the Actual System: If it is possible (cost of

to alter the system physically. For this case there is no question about whether what we

* Experiment with the model



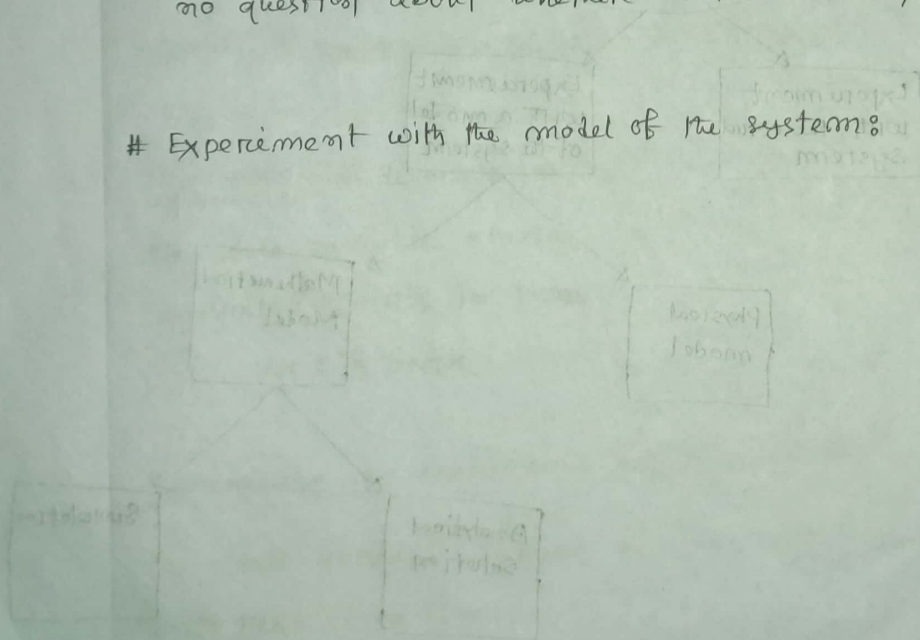
Experiment with the Actual System Vs Ex. with the model of the sys.

Experiment with the Actual system:

If it is possible (cost-effective)

to alter the system physically. For this case there is no question about whether what we study is valid.

Experiment with the model of the system:



Different ways in which a system might be studied

Physical Model Vs Mathematical Model :

Physical Model :

Physical Model also called iconic model.

A physical Model is a tangible representation of an system or object.

Ex: clay car, cockpits for airplanes

Mathematical Model: A set of mathematical equations, formulas, or algorithms that describe the behaviour of a system.

Analytic Solution vs Simulation.

Physical model is a representation of a physical system.

It is a mathematical model of a physical system.

It is a mathematical model of a physical system.

It is a mathematical model of a physical system.

It is a mathematical model of a physical system.

It is a mathematical model of a physical system.

It is a mathematical model of a physical system.

It is a mathematical model of a physical system.

Static vs Dynamic Simulation Models:

A static simulation model is a representation of a system at a particular time

Ex: Monte Carlo models

Dynamic simulation model represents a system as it ~~is~~ evolves over time

Ex: conveyor system in a factory.

Discrete event simulation:

Concerns the modeling of a system as it evolves over time by a representation in which the state variables change instantly at separate points of time.

Simulation clock: The current value of simulated time.

Next event time advance mechanism:

1. Initialize simulation clock to 0 ($t_0 = 0$)
2. Determine times of occurrence of further events from event list.
3. clock advances to next event which is executed.
4. continue until stopping rule is satisfied
5. clock "jumps" from one event time to the next. period of ~~inter~~ inactivity are ignored.

ST
↓
ABCDE

t_i = Time arrival of the i th customer ($t_0 = 0$)

$A_i = t_i - t_{i-1}$ = interarrival time

S_i = Time that server actually serving i th customer.

D_i = Delay in queue of i th customer

$e_i = t_i + D_i + S_i$ = time that i customer complete service and departs.

e_i = time of occurrence of i th event.

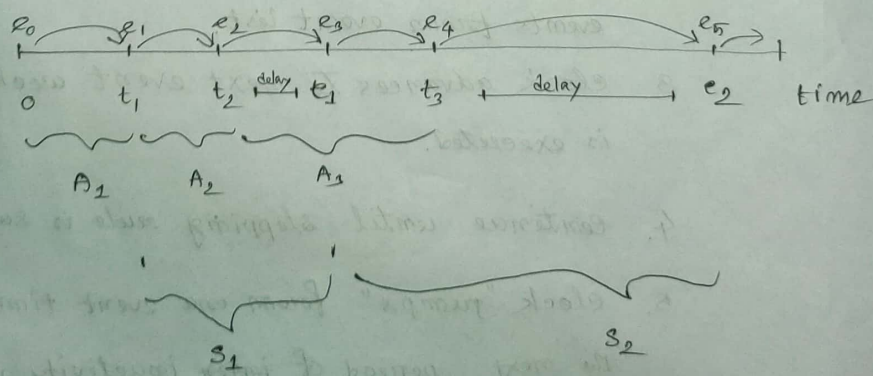


Fig: The next event time advance approach illustrated for the single server queueing system.

Component and organization of discrete-event simulation model:

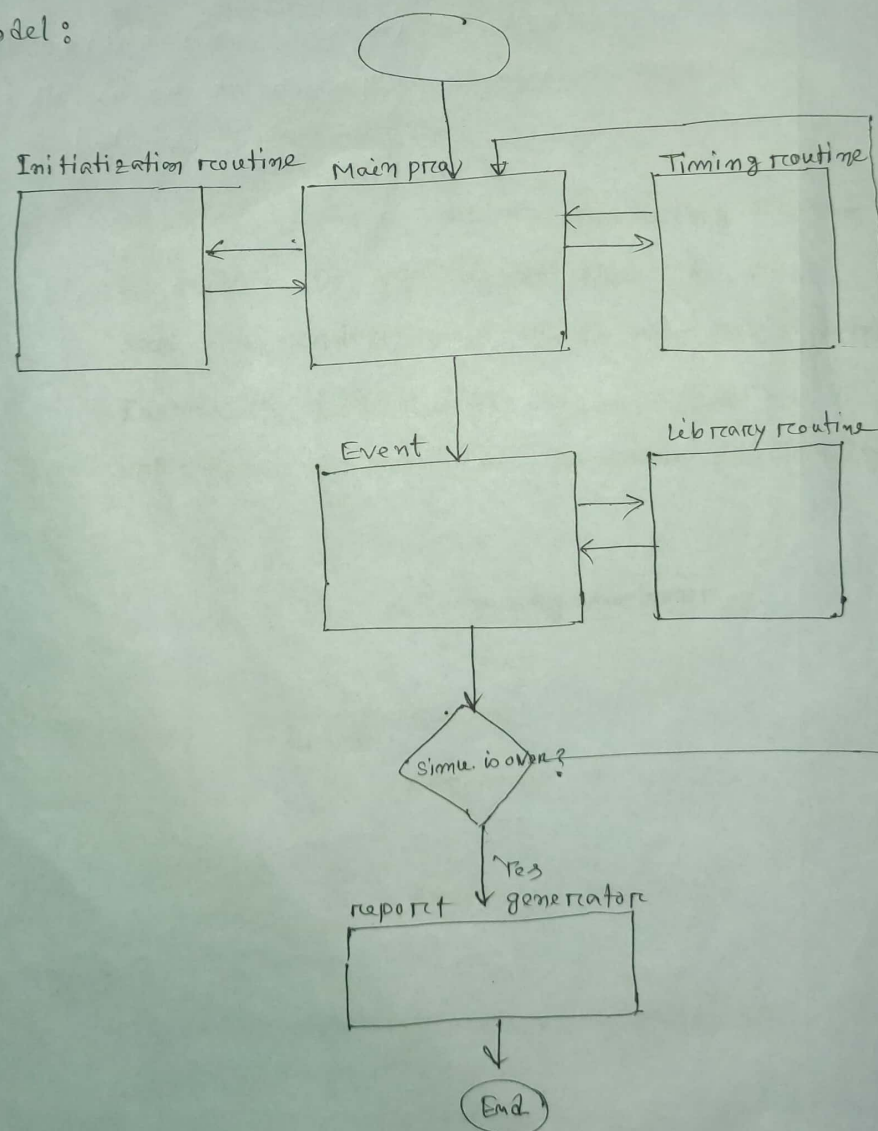
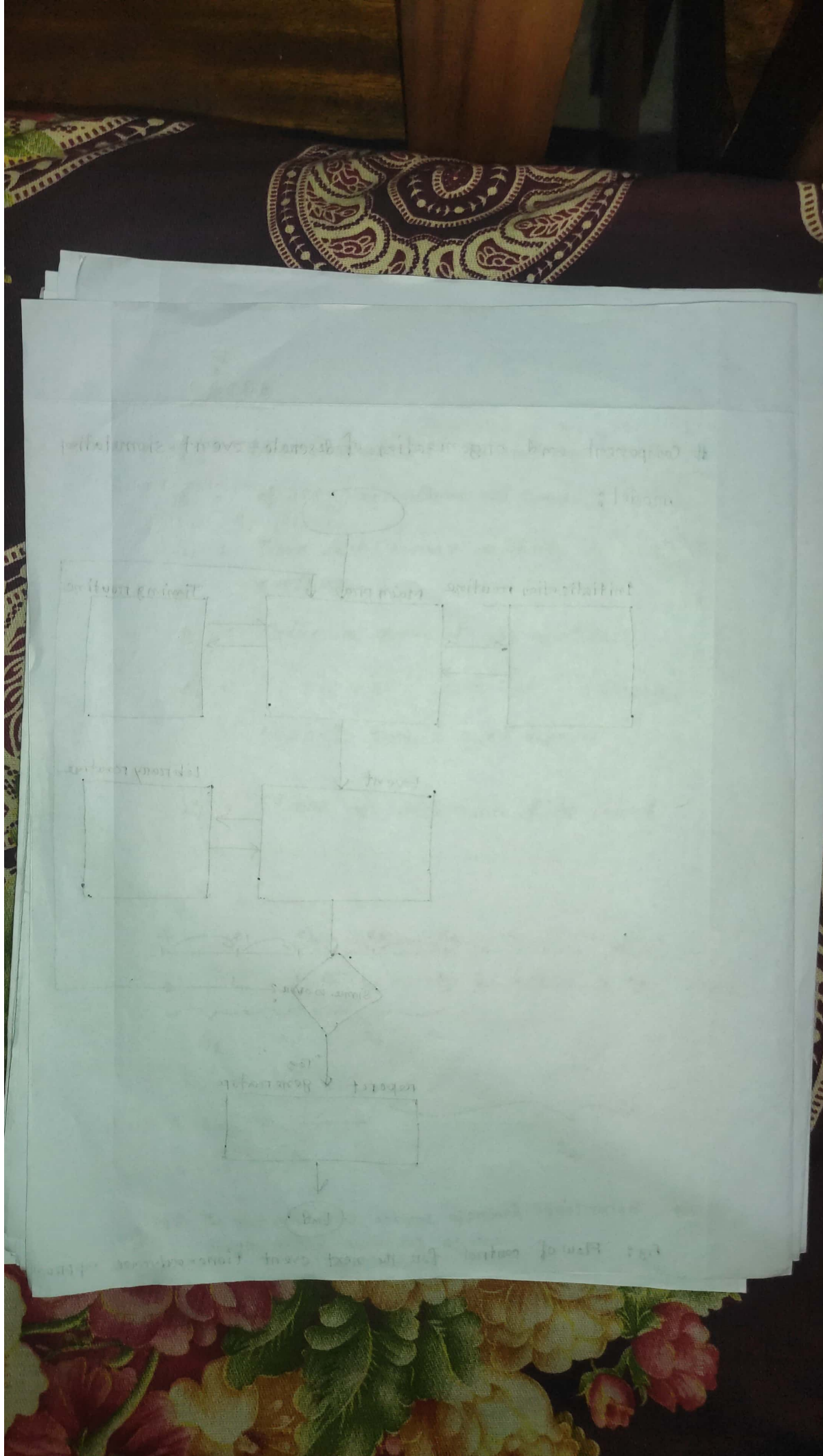


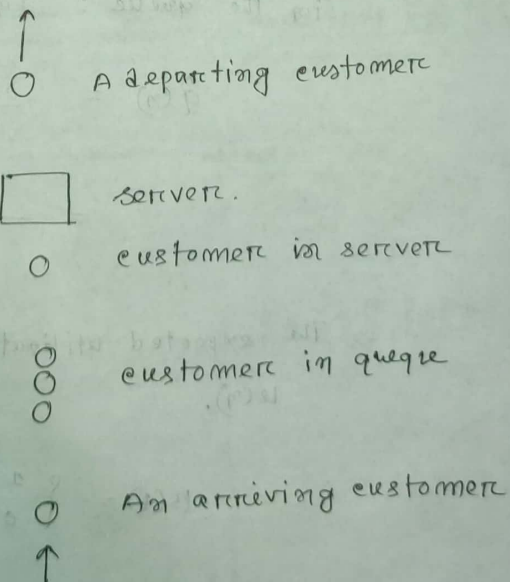
Fig: Flow of control for the next event time-advance approach



Simulation of a single server queueing system:
one-operator barbershop.

1. problem statements

Consider a single server queueing system for which the interarrival time $A_1, A_2, \dots$ are independent and identically distributed (IID).
Identically distributed means that the interarrival times have the same probability.



Service for a customer, the server choose a customer from the queue in a First-in, First-out manner

To measure the performance of the system we will estimate three quantities

1. Expected average delay in queue. ($d(m)$)

$$\hat{d}(m) = \frac{\sum_{i=1}^m D_i}{m}$$

2. Expected average number of customers in the queue: ($q(m)$)

$$\hat{q}(m) = \sum_{i=0}^{\infty} i \hat{p}_i \quad P_i = \text{expected proportion}$$

$$= \frac{\sum_{i=0}^{\infty} i T_i}{T(m)} \quad \hat{P}_i = \frac{T_i}{T(m)}$$

3. The expected utilization: How busy the server is? ($u(m)$).

$$B(t) = \begin{cases} 1 & \text{if the server is busy at time } t \\ 0 & \text{if the server is idle at time } t \end{cases}$$

$$\hat{u}(m) = \frac{\int_0^{T(m)} B(t) dt}{T(m)}$$

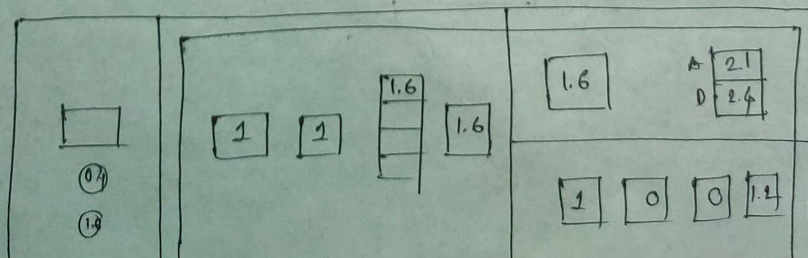
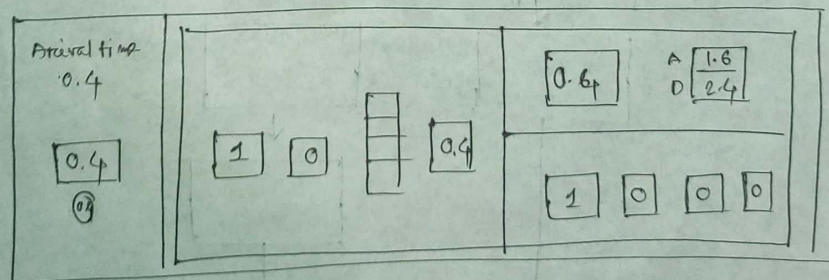
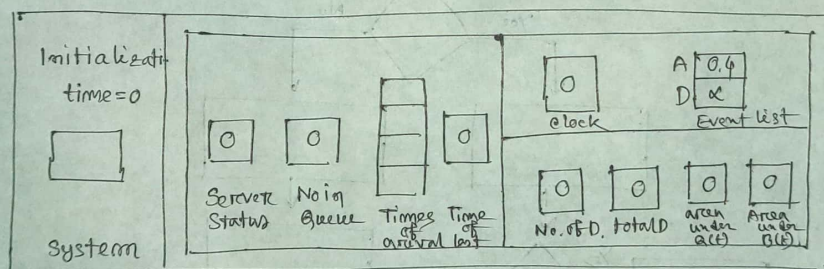
Intuitive Explanation:

Given Interval time

$$A_1 = 0.4, A_2 = 1.2, A_3 = 0.5 \dots$$

Service time

$$S_1 = 2, S_2 = 0.7, S_3 = 0.2 \dots$$



Programming organization and Logic:

Flowchart for arrival routine, queueing model:

